

Expressiveness of LDNA (reading copy)

1 Setup

We fix notation and state the two background facts the proofs rely on. An attributed graph is $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with node-feature matrix $\mathbf{X}$, adjacency matrix $\mathbf{A}$, and optional edge-feature matrix $\mathbf{E}$; row $\mathbf{x}_u$ is the feature of node $u \in \mathcal{V}$, and $\mathbf{e}_{v,u}$ is the feature of edge $(v, u) \in \mathcal{E}$. We write $\mathcal{N}_u = \{v \mid (v, u) \in \mathcal{E}\}$ for the neighbourhood of u and $n = |\mathcal{V}|$. A message-passing GNN (MP-GNN) [Gilmer et al., 2017] carries node embeddings $\mathbf{h}_u^{(l)}$, updates them by combining each node with a *permutation-invariant* aggregate of its neighbour messages, and finally reads out the multiset $\{\{\mathbf{h}_u^{(L)} \mid u \in \mathcal{V}\}\}$ into a graph embedding $\mathbf{h}_{\mathcal{G}}$.

Throughout, node and edge feature alphabets are finite and graph sizes are bounded by $n \leq N$. All expressiveness statements assert the existence of parameters realising a behaviour, in the sense of Definition 1.

The reference test. The 1-WL colour refinement (with edge colours) assigns

$$c_u^{(0)} = \mathbf{x}_u, \quad c_u^{(l+1)} = \text{HASH}\left(c_u^{(l)}, \{\{c_v^{(l)} : v \in \mathcal{N}_u\}\}\right), \quad (1)$$

with HASH injective and colours ranging over a finite set at each round; for the edge-coloured variant the neighbour term is $\{\{(c_v^{(l)}, \mathbf{e}_{v,u}) : v \in \mathcal{N}_u\}\}$. Two graphs are *1-WL-indistinguishable* if their colour histograms $\{\{c_u^{(l)} \mid u \in \mathcal{V}\}\}$ agree at every round l ; this forces equal node counts.

Definition 1 (Distinguishing power). A model class $\mathcal{M}$ (ranging over its parameters) *distinguishes* a pair $(\mathcal{G}_1, \mathcal{G}_2)$ if some parameters give $\mathcal{M}(\mathcal{G}_1) \neq \mathcal{M}(\mathcal{G}_2)$. It is *at least as expressive as* 1-WL if it distinguishes every pair that 1-WL distinguishes, and *strictly more expressive* than 1-WL if it additionally distinguishes some pair that 1-WL does not.

Fact 1 (Finite-domain realisability). Let the relevant feature and colour sets be finite.

- (a) A learnable family such as an MLP can realise any prescribed function on a finite domain.
- (b) There is an encoding of a finite set whose summation over any multiset of bounded size is injective in that multiset.

We use these only as stated; Xu et al. [2019, Lemma 5] and Zaheer et al. [2017] establish (b), and (a) is standard.

1.1 The LDNA aggregation, made precise

LDNA replaces the permutation-invariant aggregate of an MP-GNN by a learnable map applied to the neighbour messages after they are placed into fixed slots by a canonical order. We write it with three learnable functions — a message map ϕ_{Θ} , an aggregation map ψ_{θ} and an update map γ_{Θ} (all realisable as MLPs, with parameters Θ, θ) — and a canonical sorting permutation $\pi^* : \mathcal{V} \rightarrow \{1, \dots, n\}$.

Each directed edge (v, u) carries a message

$$\mathbf{m}_{v,u}^{(l+1)} = \begin{cases} \phi_{\Theta}^{(l+1)}(\mathbf{h}_u^{(l)}, \mathbf{h}_v^{(l)}) & \text{(LDNA),} \\ \phi_{\Theta}^{(l+1)}(\mathbf{h}_u^{(l)}, \mathbf{h}_v^{(l)}, \mathbf{e}_{v,u}) & \text{(LDNA-e).} \end{cases} \quad (2)$$

The messages of u are gathered into a fixed-size matrix $\mathbf{S}_u^{(l+1)} \in \mathbb{R}^{N \times d_h}$ by scattering each neighbour's message to the row indexed by its canonical position: for $p = 1, \dots, N$,

$$(\mathbf{S}_u^{(l+1)})_p = \sum_{v \in \mathcal{N}_u} \mathbb{I}[\pi^*(v) = p] \mathbf{m}_{v,u}^{(l+1)}. \quad (3)$$

Because π^* is injective, each row receives at most one message; rows whose position is not that of a neighbour are $\mathbf{0}$. The row index is the neighbour's *global* canonical position, so reading a row recovers which node contributed it — the identity information that a permutation-invariant aggregate discards. The node is then updated from its own state and the aggregated slots, and after L layers a readout pools the node embeddings:

$$\mathbf{h}_u^{(l+1)} = \gamma_{\Theta}^{(l+1)}(\mathbf{h}_u^{(l)}, \psi_{\Theta}^{(l+1)}(\mathbf{S}_u^{(l+1)})), \quad (4)$$

$$\mathbf{h}_{\mathcal{G}} = \text{READOUT}(\{\{\mathbf{h}_u^{(L)} \mid u \in \mathcal{V}\}\}). \quad (5)$$

The update (4) keeps $\mathbf{h}_u^{(l)}$ as an explicit self-argument.

Definition 2 (Canonical labelling). A *canonical labelling* assigns to each graph $\mathcal{G}$ a bijection $\pi_{\mathcal{G}}^* : \mathcal{V} \rightarrow \{1, \dots, n\}$ that is isomorphism-invariant at the labelled-graph level: for every isomorphism $f : \mathcal{G}_1 \rightarrow \mathcal{G}_2$, relabelling by $\pi_{\mathcal{G}_1}^*$ and $\pi_{\mathcal{G}_2}^*$ yields the same ordered attributed graph. Such a labelling is obtainable by completing a feature-based sort through individualization–refinement [McKay and Piperno, 2014], with quasi-polynomial worst-case cost [Babai, 2016].

2 LDNA is strictly more expressive than 1-WL

Theorem 1. *If π^* is induced by a canonical labelling (Definition 2), then LDNA (Equations (2) to (5)) is strictly more expressive than the 1-WL test, and hence than any MP-GNN with permutation-invariant aggregation [Xu et al., 2019, Morris et al., 2019]. The same holds for LDNA-e against the edge-coloured 1-WL refinement.*

We prove the theorem through two lemmas: LDNA can simulate 1-WL (Lemma 1), and LDNA separates a 1-WL-indistinguishable pair (Lemma 2).

Lemma 1 (Simulation). *LDNA is at least as expressive as 1-WL.*

Proof. Claim. There exist parameters and, for each layer l , an injective map α_l such that $\mathbf{h}_u^{(l)} = \alpha_l(c_u^{(l)})$ for every node u of every graph of size $\leq N$.

Base ($l = 0$). $c_u^{(0)} = \mathbf{x}_u$, and taking the input encoding to be injective on the finite feature alphabet (Fact 1(a)) gives $\mathbf{h}_u^{(0)} = \alpha_0(\mathbf{x}_u)$ with α_0 injective.

Inductive step. Assume the claim at layer l ; we realise it at layer $l + 1$.

(1) *Message.* Since α_l is injective and all arguments range over finite sets, $\phi_{\Theta}^{(l+1)}$ can encode the pair $(\mathbf{h}_v^{(l)}, \mathbf{e}_{v,u})$ — equivalently $(c_v^{(l)}, \mathbf{e}_{v,u})$ — injectively into $\mathbf{m}_{v,u}^{(l+1)}$ (Fact 1(a)).

(2) *Aggregation.* Let $\psi_{\Theta}^{(l+1)}$ ignore the slot positions and sum the rows of $\mathbf{S}_u^{(l+1)}$; by (3) this equals $\sum_{v \in \mathcal{N}_u} \mathbf{m}_{v,u}^{(l+1)}$. With the message encoding of step (1) chosen summation-injective (Fact 1(b)), the value $\psi_{\Theta}^{(l+1)}(\mathbf{S}_u^{(l+1)})$ is then an injective function of the neighbour multiset

$$\{(c_v^{(l)}, \mathbf{e}_{v,u}) \mid v \in \mathcal{N}_u\}.$$

(3) *Update.* The self-argument $\mathbf{h}_u^{(l)}$ of $\gamma_{\Theta}^{(l+1)}$ lets it injectively combine $c_u^{(l)}$ with the aggregate (Fact 1(a)); an empty neighbourhood is harmless: the empty sum is, by Fact 1(b), distinct from every non-empty aggregate, and the self-argument determines the output.

Composing (1)–(3), $\mathbf{h}_u^{(l+1)}$ is an injective function of $(c_u^{(l)}, \{(c_v^{(l)} \mid v \in \mathcal{N}_u\})$, which by injectivity of HASH in (1) determines $c_u^{(l+1)}$; set α_{l+1} accordingly, proving the claim.

Graph level. Take the readout to sum a summation-injective encoding of the final embeddings (Fact 1(b)); by the claim it is injective in the colour histogram $\{\{c_u^{(L)}\}\}$. Hence whenever 1-WL assigns two graphs different colour histograms at some round, LDNA (run to that depth) assigns them different graph embeddings. The edge-coloured variant is identical with $\mathbf{e}_{v,u}$ carried through step (1). $\square$

Lemma 2 (Separation). *There is a 1-WL-indistinguishable pair of graphs that LDNA distinguishes.*

Proof. Fix $m \geq 3$ and let $\mathcal{G}_1 = C_{2m}$ be the cycle on $2m$ nodes and $\mathcal{G}_2 = C_m \sqcup C_m$ the disjoint union of two m -cycles, both with a single constant node feature and no edge features.

Step 1: the pair is 1-WL-indistinguishable. Both graphs are 2-regular on $2m$ nodes with identical initial colours. By induction on l , all nodes share one colour at every round: if every node has colour a at round l , each has neighbour multiset $\{\{a, a\}\}$, so (1) returns a single colour at round $l + 1$. The colour histograms therefore coincide at every round, so $\mathcal{G}_1$ and $\mathcal{G}_2$ are 1-WL-indistinguishable.

Step 2: the canonical labelling. A canonizer may fix any one representative labelling per isomorphism class. For C_{2m} take the cyclic labelling on $\{1, \dots, 2m\}$, so that position p is adjacent to positions $p \pm 1$ modulo $2m$. For $C_m \sqcup C_m$ take the two consecutive blocks

$$B_1 = \{1, \dots, m\}, \quad B_2 = \{m + 1, \dots, 2m\},$$

each of the two m -cycles cyclically labelled within its own block.

Step 3: separating parameters, algebraically. Take $\phi_{\Theta} \equiv \mathbf{1}$ on a single channel, so every message equals 1 and, by (3), $\mathbf{S}_u$ is the indicator vector of the *slot set*

$$P_u := \{\pi^*(v) \mid v \in \mathcal{N}_u\} \subseteq \{1, \dots, 2m\}.$$

Now compare the two labellings. In $\mathcal{G}_2$ every edge joins two nodes of the same block, so every P_u lies inside a single block. In $\mathcal{G}_1$ the node at position 1 has neighbours at positions 2 and $2m$, giving

$$P = \{2, 2m\}, \quad 2 \in B_1, \quad 2m \in B_2,$$

using $2 \leq m$ and $2m \geq m + 1$ (both hold for $m \geq 3$); this P meets both blocks. Consequently the multiset $\{\{P_u \mid u \in \mathcal{V}\}\}$ of $\mathcal{G}_1$ contains a set contained in no single block, whereas that of $\mathcal{G}_2$ contains none, so the two multisets differ. Let ψ_{Θ} realise an injective map on the finite domain of indicator vectors (Fact 1(a)); then $\psi_{\Theta}(\mathbf{S}_u)$ is injective in P_u . Let γ_{Θ} be injective; since $\mathbf{h}_u^{(0)}$ is the same constant in both graphs, one layer yields $\mathbf{h}_u^{(1)}$ as an injective function of P_u , so $\{\{\mathbf{h}_u^{(1)}\}\}$ is the image of $\{\{P_u\}\}$ under an injective map — and these image multisets differ across $\mathcal{G}_1$ and $\mathcal{G}_2$.

Step 4: conclusion. Taking the readout injective (Fact 1(b)), the two graphs receive different embeddings; since 1-WL cannot separate them, LDNA distinguishes a 1-WL-indistinguishable pair. The smallest instance is $m = 3$: C_6 versus two triangles. $\square$

Proof of Theorem 1. Lemma 1 gives $\text{LDNA} \succeq 1\text{-WL}$ and Lemma 2 gives a 1-WL-indistinguishable pair that LDNA separates; by Definition 1 LDNA is strictly more expressive than 1-WL, and hence than any MP-GNN with permutation-invariant aggregation, whose power is bounded by 1-WL [Xu et al., 2019, Morris et al., 2019]. The LDNA-e statement follows from the same two lemmas against the edge-coloured refinement. $\square$

Proposition 1 (The canonical labelling is necessary). *The separation in Theorem 1 depends on π^* being a canonical labelling.*

- (i) *If the stacked input (3) is replaced by the content-sorted list of the messages (ties left unbroken), then LDNA is bounded by 1-WL, and by Lemma 1 is exactly 1-WL.*
- (ii) *If π^* is obtained by sorting on isomorphism-invariant per-node quantities, it individualizes no node of a vertex-transitive graph, so on the classical 1-WL-hard pairs it supplies no identities.*

Proof. (i). Sorting the messages by content is a bijection from the multiset $\{\{\mathbf{m}_{v,u} \mid v \in \mathcal{N}_u\}\}$ to a sorted tuple, so $\psi_\theta \circ \text{sort}$ is a function of that multiset alone, i.e. a permutation-invariant aggregate. LDNA is then an MP-GNN with permutation-invariant aggregation, whose power is bounded by 1-WL [Xu et al., 2019, Morris et al., 2019]; with Lemma 1 — whose aggregation is the order-agnostic summation, hence available under content-sorting unchanged — it is exactly 1-WL.

(ii). Let ι be an isomorphism-invariant node quantity (raw features, degree, 1-WL colour, subgraph counts, and the like), and let π^* sort by ι . If $\mathcal{G}$ is vertex-transitive, then for any two nodes u, w some automorphism maps u to w , whence $\iota(u) = \iota(w)$; thus ι assigns every node the same sort key, so it induces no individualizing invariant order, and any bijection π^* extending it is determined entirely by tie-breaking. On a vertex-transitive 1-WL-indistinguishable pair — the 4×4 rook’s graph versus the Shrikhande graph, or the CSL family — such an order therefore supplies no identities, and the construction of Lemma 2 is unavailable. Breaking the ties arbitrarily makes the output depend on the input labelling, destroying well-definedness. A canonical labelling (Definition 2) individualizes the nodes while remaining invariant at the labelled-graph level, and is the only way to secure both. $\square$

Remark 1 (Random initialization). With the parameters of $\phi_\Theta, \psi_\theta, \gamma_\Theta$ drawn independently from a continuous distribution, each layer’s maps are injective almost surely: over the finite domain of colours, the parameter settings that collapse two distinct colours form a finite union of measure-zero sets. Composing, the embeddings $\mathbf{h}_u^{(l)}$ refine the 1-WL colours $c_u^{(l)}$ almost surely, and the same genericity makes the read-out injective on the refined histograms, so a randomly initialized LDNA is at least as expressive as 1-WL without any training. The strict separation of Lemma 2 instead uses the explicit deterministic parameters constructed there. This differs from random node initialization [Abboud et al., 2021], which randomizes the inputs rather than the aggregator.

Remark 2 (Permutation invariance). Although π^* imposes an order, the LDNA output is a deterministic graph invariant: by Definition 2 the labelling depends only on the isomorphism class, and any choice of representative within an automorphism orbit yields the same labelled graph — hence the same stacked inputs $\mathbf{S}_u$ and the same embeddings. The sort that secures permutation invariance and the labelling that supplies node identities are one and the same object.

References

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